

```
PS C:\Users\NEHA\OneDrive\Desktop\Navodita Internship\Navodita_Infotech_Internship_tasks\DSA Visualizer> g++ main.cpp Sorting.cpp Utils.cpp -o visualizer
PS C:\Users\NEHA\OneDrive\Desktop\Navodita Internship\Navodita_Infotech_Internship_tasks\DSA Visualizer> .\visualizer.exe
```

```
=====
DATA STRUCTURES & ALGORITHMS VISUALIZER
=====
```

Current Array:

7 3 15 12 1 13 7 13 17 11

```
7 | *****
3 | ***
15 | *****
12 | *****
1 | *
13 | *****
7 | *****
13 | *****
17 | *****
11 | *****
```

Welcome to the Data Structures and Algorithms Visualizer
This project demonstrates the working of different sorting algorithms.

1. Bubble Sort
2. Selection Sort
3. Insertion Sort
4. Merge Sort
5. Quick Sort
6. Generate New Array
7. About Project
8. Exit

Enter Choice : 1

===== BUBBLE SORT STARTED =====

Pass 1

```
Comparing 7 and 3
Swapped!
3 7 15 12 1 13 7 13 17 11
3 | ***
7 | *****
15 | *****
12 | *****
1 | *
13 | *****
7 | *****
13 | *****
17 | *****
11 | *****
```

```
Comparing 7 and 15
No Swap Needed.
3 7 15 12 1 13 7 13 17 11
3 | ***
7 | *****
15 | *****
12 | *****
1 | *
13 | *****
7 | *****
13 | *****
17 | *****
11 | *****
```

```
Comparing 15 and 12
Swapped!
3 7 12 15 1 13 7 13 17 11
3 | ***
7 | *****
12 | *****
15 | *****
1 | *
13 | *****
7 | *****
13 | *****
```

```
13 | *****
7  | *****
13 | *****
17 | *****
11 | *****
```

Comparing 15 and 1
Swapped!

3 7 12 1 15 13 7 13 17 11

```
3 | ***
7 | *****
12 | *****
1 | *
15 | *****
13 | *****
7  | *****
13 | *****
17 | *****
11 | *****
```

Comparing 15 and 13
Swapped!

3 7 12 1 13 15 7 13 17 11

```
3 | ***
7 | *****
12 | *****
1 | *
13 | *****
15 | *****
7  | *****
13 | *****
17 | *****
11 | *****
```

```
Comparing 15 and 7
Swapped!
3 7 12 1 13 7 15 13 17 11
3 | ***
7 | *****
12 | *****
1 | *
13 | *****
7 | *****
15 | *****
13 | *****
17 | *****
11 | *****
```

```
Comparing 15 and 13
Swapped!
3 7 12 1 13 7 13 15 17 11
3 | ***
7 | *****
12 | *****
1 | *
13 | *****
7 | *****
13 | *****
15 | *****
17 | *****
11 | *****
```

```
Comparing 15 and 17
No Swap Needed.
3 7 12 1 13 7 13 15 17 11
3 | ***
7 | *****
12 | *****
1 | *
13 | *****
7 | *****
13 | *****
15 | *****
```

```
15 | *****
17 | *****
11 | *****
```

Comparing 17 and 11
Swapped!

```
3 7 12 1 13 7 13 15 11 17
3 | ***
7 | *****
12 | *****
1 | *
13 | *****
7 | *****
13 | *****
15 | *****
11 | *****
17 | *****
```

Pass 2

Comparing 3 and 7
No Swap Needed.

```
3 7 12 1 13 7 13 15 11 17
3 | ***
7 | *****
12 | *****
1 | *
13 | *****
7 | *****
13 | *****
15 | *****
11 | *****
17 | *****
```

```
Comparing 7 and 12
No Swap Needed.
3 7 12 1 13 7 13 15 11 17
3 | ***
7 | *****
12 | *****
1 | *
13 | *****
7 | *****
13 | *****
15 | *****
11 | *****
17 | *****
```

```
Comparing 12 and 1
Swapped!
3 7 1 12 13 7 13 15 11 17
3 | ***
7 | *****
1 | *
12 | *****
13 | *****
7 | *****
13 | *****
15 | *****
11 | *****
17 | *****
```

```
Comparing 12 and 13
No Swap Needed.
3 7 1 12 13 7 13 15 11 17
3 | ***
7 | *****
1 | *
12 | *****
13 | *****
7 | *****
13 | *****
15 | *****
```

```
15 | *****
11 | *****
17 | *****
```

```
Comparing 13 and 7
Swapped!
3 7 1 12 7 13 13 15 11 17
```

```
3 | ***
7 | *****
1 | *
12 | *****
7 | *****
13 | *****
13 | *****
15 | *****
11 | *****
17 | *****
```

```
Comparing 13 and 13
No Swap Needed.
3 7 1 12 7 13 13 15 11 17
```

```
3 | ***
7 | *****
1 | *
12 | *****
7 | *****
13 | *****
13 | *****
15 | *****
11 | *****
17 | *****
```

```
Comparing 13 and 15
No Swap Needed.
3 7 1 12 7 13 13 15 11 17
```

```
3 | ***
7 | *****
1 | *
```

```
3 | ***
7 | *****
1 | *
12 | *****
7 | *****
13 | *****
13 | *****
15 | *****
11 | *****
17 | *****
```

Comparing 15 and 11
Swapped!

```
3 7 1 12 7 13 13 11 15 17
3 | ***
7 | *****
1 | *
12 | *****
7 | *****
13 | *****
13 | *****
11 | *****
15 | *****
17 | *****
```

Pass 3

Comparing 3 and 7
No Swap Needed.

```
3 7 1 12 7 13 13 11 15 17
3 | ***
7 | *****
1 | *
12 | *****
7 | *****
13 | *****
13 | *****
11 | *****
```



```
11 | *****
15 | *****
17 | *****
```

Comparing 7 and 1

Swapped!

```
3 1 7 12 7 13 13 11 15 17
```

```
3 | ***
1 | *
7 | *****
12 | *****
7 | *****
13 | *****
13 | *****
11 | *****
15 | *****
17 | *****
```

Comparing 7 and 12

No Swap Needed.

```
3 1 7 12 7 13 13 11 15 17
```

```
3 | ***
1 | *
7 | *****
12 | *****
7 | *****
13 | *****
13 | *****
11 | *****
15 | *****
17 | *****
```

Comparing 12 and 7

Swapped!

```
3 1 7 7 12 13 13 11 15 17
```

```
3 | ***
1 | *
7 | *****
```

Comparing 12 and 7
Swapped!

3 1 7 7 12 13 13 11 15 17
3 | ***
1 | *
7 | *****
7 | *****
12 | *****
13 | *****
13 | *****
11 | *****
15 | *****
17 | *****

Comparing 12 and 13
No Swap Needed.

3 1 7 7 12 13 13 11 15 17
3 | ***
1 | *
7 | *****
7 | *****
12 | *****
13 | *****
13 | *****
11 | *****
15 | *****
17 | *****

Comparing 13 and 13
No Swap Needed.

3 1 7 7 12 13 13 11 15 17
3 | ***
1 | *
7 | *****
7 | *****
12 | *****
13 | *****
13 | *****
11 | *****

Comparing 3 and 7
No Swap Needed.
1 3 7 7 11 12 13 13 15 17

1		*
3		***
7		*****
7		*****
11		*****
12		*****
13		*****
13		*****
15		*****
17		*****

Comparing 7 and 7
No Swap Needed.
1 3 7 7 11 12 13 13 15 17

1		*
3		***
7		*****
7		*****
11		*****
12		*****
13		*****
13		*****
15		*****
17		*****

Comparing 7 and 11
No Swap Needed.
1 3 7 7 11 12 13 13 15 17

1		*
3		***
7		*****
7		*****
11		*****
12		*****
13		*****
13		*****

```
13 | *****
15 | *****
17 | *****
```

Pass 7

```
Comparing 1 and 3
No Swap Needed.
1 3 7 7 11 12 13 13 15 17
1 | *
3 | ***
7 | *****
7 | *****
11 | *****
12 | *****
13 | *****
13 | *****
15 | *****
17 | *****
```

```
Comparing 3 and 7
No Swap Needed.
1 3 7 7 11 12 13 13 15 17
1 | *
3 | ***
7 | *****
7 | *****
11 | *****
12 | *****
13 | *****
13 | *****
15 | *****
17 | *****
```

```
Comparing 7 and 7
No Swap Needed.
1 3 7 7 11 12 13 13 15 17
```

1 3 7 7 11 12 13 13 15 17

```
1 | *
3 | ***
7 | *****
7 | *****
11 | *****
12 | *****
13 | *****
13 | *****
15 | *****
17 | *****
```

Pass 8

Comparing 1 and 3

No Swap Needed.

1 3 7 7 11 12 13 13 15 17

```
1 | *
3 | ***
7 | *****
7 | *****
11 | *****
12 | *****
13 | *****
13 | *****
15 | *****
17 | *****
```

Comparing 3 and 7

No Swap Needed.

1 3 7 7 11 12 13 13 15 17

```
1 | *
3 | ***
7 | *****
7 | *****
11 | *****
12 | *****
13 | *****
```

```
13 | *****
13 | *****
15 | *****
17 | *****
```

Pass 9

Comparing 1 and 3

No Swap Needed.

1 3 7 7 11 12 13 13 15 17

```
1 | *
3 | ***
7 | *****
7 | *****
11 | *****
12 | *****
13 | *****
13 | *****
15 | *****
17 | *****
```

=====

Bubble Sort Completed!

=====

Sorted Array:

1 3 7 7 11 12 13 13 15 17

Statistics

Comparisons : 45

Swaps : 16

Time Complexity

Best Case : $O(n)$

Average Case : $O(n^2)$

Worst Case : $O(n^2)$

Space Complexity : $O(1)$